Mass Storage Devices - HDD and SSD

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1 Mass Storage Devices

Mass storage devices are storage devices used for the permanent storage of files.

Mass storage devices contain:

- Our **operating system**;
- The **file system**, that is, the entire hierarchy of folders containing the operating system files and our personal files;
- All our programs, images, videos, photos, games, and so on.

Mass storage devices can be:

- **Optical**: CD, DVD;
- **Mechanical**: HDD (Hard Disk Drive);
- **Electronic**: SSD (Solid State Drive).

2 Difference Between HDD and SSD

Let's focus on the difference between HDD and SSD.

The main difference between the two technologies concerns the way data is read and written.

2.1 SSD (Solid State Drive)

An SSD consists of billions of "electrical cells" (NAND gates, essentially made up of a pair of transistors), each of which can be electrically charged (**1**) or uncharged (**0**).

2.2 HDD (Hard Disk Drive)

A hard disk, on the other hand, uses a **magnetic read/write head** capable of reading the content stored on the "platters", that is, metallic disks.

We can imagine them as small CDs stacked on top of one another, which can spin at speeds of up to 7000 revolutions per minute.

2.3 HDD vs SSD

An SSD is much faster (up to four times faster than an HDD), thanks to the electronic nature of the device, compared to the mechanical nature of an HDD.

However, it has some disadvantages:

- It is generally more expensive than an HDD;
- It takes up less physical space than HDDs, especially in the case of NVMe SSDs.

Another important difference concerns data retention.

An HDD allows data to be stored for a much longer time compared to an SSD. The latter, in fact, is subject to a phenomenon called **charge decay**, which consists of the gradual loss of charge in the electrical cells responsible for representing "1" in stored data.

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[1] Parts of this document are LLM-generated output.